

Latin Square Designs

Example: Consider an experiment about the effects of five types of background music (A,B,C,D,E) on the productivity of bank tellers. Day of the week (Monday through Friday) and week of the experimental period are to be the two blocking variables. Suppose we select p weeks. If for the above experiment we want to use a rcbd design then we will have to have (if no replication is employed): $5 \times 5 \times p$ bank tellers in the experiment. If p is moderate it will amount to too many experimental units (tellers). In many instances, in particular in pilot studies, one has to resort to an incomplete rbd (irbd) design and the most economical and simple one is the *Latin square design*.

In this design each treatment appears once and only once for each of the two blocking variables. In this design we are still interested in the treatments only. It is used to eliminate two sources of variability, and we have two restriction on randomization. In this design we have to have the number of treatments equal to the number of levels of both restrictions on randomization. A Latin square for p treatment levels, or a $p \times p$ Latin square, is a square containing p rows and p columns.

Question: Is the randomization completely lost in this design?

The answer is no! There are $p!(p-1)!v$ different $p \times p$ Latin squares, where v is the number of standard squares, where a standard square is a square in which the first row and column is written in alphabetical order.

Standard Latin Squares and Number of Latin Square $p=1, \dots, 7$

Size	3x3	4x4	5x5	6x6	7x7
Standard Squares	1	4	56	9,408	16,942,080
Total Squares	12	576	161,280	818,851,200	61,479,419,904,000

Remark: Little is known about the properties of Latin squares bigger than 7×7 .

Advantages of Latin Square Designs

1. The use of two blocking variables often permits greater reduction in the variability of experimental errors than can be obtained with either of the blocking variables when they are employed alone.
2. Treatment effects can be studied from a small-scale experiment. This is particularly useful in preliminary or pilot studies.
3. Useful in repeated measure designs.
4. Useful in cross-over designs.

Disadvantages of Latin Square Designs

1. The number of classes of each blocking variable must equal to the number of treatments.
2. Restrictive assumptions: no interactions are allowed.
3. The two blocking variables have to have the same number of levels.
4. The randomization required is more complex.

Data for our example: (productivity of tellers, data coded)

Week	Day					Total
	M	T	W	Th	F	
1	18(D)	17(C)	14(A)	21(B)	17(E)	$Y_{1..} = 87$
2	13(C)	34(B)	21(E)	16(A)	15(D)	$Y_{2..} = 99$
3	7(A)	29(D)	32(B)	27(E)	13(C)	$Y_{3..} = 108$
4	17(E)	13(A)	24(C)	31(D)	25(B)	$Y_{4..} = 110$
5	21(B)	26(E)	26(D)	31(C)	7(A)	$Y_{5..} = 111$

Total $Y_{..1}=76$ $Y_{..2}=119$ $Y_{..3}=117$ $Y_{..4}=126$ $Y_{..5}=77$ $Y_{...}=515$

$$Y_{.1.} = 7 + 13 + 14 + 16 + 7 = 57 \quad \bar{Y}_{.1.} = 11.4$$

$$Y_{.2.} = 21 + 34 + 32 + 21 + 25 = 133 \quad \bar{Y}_{.2.} = 26.6$$

$$Y_{.3.} = 13 + 17 + 24 + 31 + 13 = 98 \quad \bar{Y}_{.3.} = 19.6$$

$$Y_{.4.} = 18 + 29 + 26 + 31 + 15 = 119 \quad \bar{Y}_{.4.} = 23.8$$

$$Y_{.5.} = 17 + 26 + 21 + 27 + 17 = 108 \quad \bar{Y}_{.5.} = 21.6$$

A Latin square design involves the main effect of the row blocking variable denoted by α_i , the main effect of the column blocking variable denoted by β_k , and the treatment main effect denoted by τ_j . It is assumed that no interaction exists between these three main effects. Thus, the model employed is an additive one and is given by:

$$Y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk},$$

where, $i=1,\dots,p$, $j=1,\dots,p$, $k=1,\dots,p$ and

μ is a constant, the overall mean,

α_i are constants and sum up to 0 if the row effect is fixed, and are iid $N(0, \sigma^2_\alpha)$ if it is random,

β_k are constants and sum up to 0 if the column effect is fixed, and are iid $N(0, \sigma^2_\beta)$ if it is random,

τ_j are constants and sum up to 0 if the treatment effect is fixed, and are iid $N(0, \sigma^2_\tau)$ if it is random,

ε_{ijk} are iid $N(0, \sigma^2_\epsilon)$ (and independent of other effects if they are random).

Fitted Model

The least square estimators of the parameters in Latin square model are:

Parameter Estimator

$$\mu \quad \bar{Y}_{...}$$

$$\alpha_i \quad \bar{Y}_{i..} - \bar{Y}_{...}$$

$$\beta_k \quad \bar{Y}_{..k} - \bar{Y}_{...}$$

$$\tau_j \quad \bar{Y}_{.j.} - \bar{Y}_{...}$$

The fitted values are:

$$\hat{Y}_{ijk} = \bar{Y}_{i..} + \bar{Y}_{.j.} + \bar{Y}_{..k} - 2\bar{Y}_{...}$$

and the residuals are:

$$e_{ijk} = Y_{ijk} - \hat{Y}_{ijk} = Y_{ijk} - \bar{Y}_{i..} - \bar{Y}_{.j.} - \bar{Y}_{..k} + 2\bar{Y}_{...}$$

We now proceed to discuss:

The ANOVA for the Latin Square design and examine the SAS output for our data set.

Efficiency Measures for the Blocking Variables

The efficiency measure for the effectiveness of Latin square design (lsd) vs. a completely randomized design (crd):

$$E_1 = \sigma_{\text{crd}}^2 / \sigma_{\text{lsd}}^2$$

which is estimated by

$$\hat{E}_1 = \hat{\sigma}_{\text{crd}}^2 / \hat{\sigma}_{\text{lsd}}^2$$

where

$$\sigma_{\text{lsd}}^2 = MS_{\text{Error}}$$

$\hat{\sigma}_{\text{crd}}^2 =$

$$[MS_{\text{row}} + MS_{\text{column}} + (p-1)MS_{\text{Error}}] / (p+1).$$

Therefore, for our example we get

$\hat{E}_1 =$

$$\begin{aligned} \hat{E}_1 &= [MS_{\text{Week}} + MS_{\text{Day}} + 4MS_{\text{Error}}] / [6MS_{\text{Error}}] \\ &= [20.5 + 119.3 + 4(15.7)] / [6(15.7)] = 2.2 \end{aligned}$$

The efficiency measure for the effectiveness of a Latin square resign (lsd) vs. a randomized block design with the row blocking variable

$$E_2 = \sigma^2_{\text{crbdrow}} / \sigma^2_{\text{lsd}}$$

which is estimated by

$$\hat{E}_2 = \hat{\sigma}^2_{\text{crbdrow}} / \hat{\sigma}^2_{\text{lsd}}$$

where

$$\hat{\sigma}^2_{\text{crbdrow}} = [\text{MS}_{\text{column}} + (p-1)\text{MS}_{\text{Error}}]/p.$$

For our data set Day was the row variable and we get

$$\hat{E}_2 = [20.5 + 4(15.7)]/[5(15.7)] = 1.1$$

This implies that the rbd with day as a blocking variable is as efficient as the lsd design.

The efficiency measure for the effectiveness of Latin square resign (lsd) vs. a randomized block design with the column variable:

$$E_3 = \sigma^2_{\text{crbdcolum}} / \sigma^2_{\text{lsd}}$$

which is estimated by

$$\hat{E}_3 = \hat{\sigma}_{\text{crbdcolum}}^2 / \hat{\sigma}_{\text{lsd}}^2$$

where

$$\hat{\sigma}_{\text{crbdcolum}}^2 = [\text{MS}_{\text{row}} + (p-1)\text{MS}_{\text{Error}}]/p.$$

For our example (where **week** was the column variable) we get

$$\begin{aligned}\hat{E}_3 &= [\text{MS}_{\text{Day}} + 4\text{MS}_{\text{Error}}]/[5\text{MS}_{\text{Error}}] \\ &= [119.3 + 4(15.7)]/[5(15.7)] = 2.3\end{aligned}$$

This implies that the Latin square design is 2.3 times more efficient than the rbd design with the variable week as a blocking variable.

Graeco-Latin Square Design

Consider a $p \times p$ Latin square, and superimposed on it a second Latin square in which the levels of the third blocking variable are denoted by lower case Greek letters: $\alpha, \beta, \gamma, \delta, \epsilon$, and so on. If each of these Greek letters appears only once with each of the Latin letters, then the two squares are said to be orthogonal and the design is called a *Graeco-Latin square design*.

Example : Let us consider again the design for the five types of music data and its relation to productivity. Suppose there are also five periods of the day during which the music is played:

1 - whole day	α
2 - morning 4 hours	β
3 - afternoon 4 hours	γ
4 - 2 hours in the morning and 2 in the afternoon	δ
5 - 2 hours in the afternoon	ϵ

Data:

Week	Monday	Tuesday	Wednesday	Thursday	Friday
1	18(D β)	17(E γ)	14(A α)	21(B ϵ)	17(C δ)
2	17(E ϵ)	13(C α)	15(D δ)	16(A γ)	34(B δ)
3	13(C γ)	32(B δ)	27(E β)	29(D α)	7(A ϵ)
4	13(A δ)	31(D ϵ)	25(B γ)	24(C β)	21(E α)
5	21(B α)	7(A β)	31(C ϵ)	21(E δ)	26(D γ)

Model:
$$Y_{ijkl} = \mu + \theta_i + \tau_j + \omega_k + \Psi_l + \varepsilon_{ijkl}$$

where Ψ_l is the extra blocking random variable with levels

$\alpha, \beta, \gamma, \delta, \epsilon$, and $i=1, \dots, p, j=1, \dots, p, k=1, \dots, p, l=1, \dots, p$ and μ is a constant, the overall mean,

θ_i are constants and sum up to 0 if the row effect is fixed,
and are iid $N(0, \sigma_\theta^2)$ if it is random,

τ_j are constants and sum up to 0 if the effect of the Latin letter treatment is fixed and are iid $N(0, \sigma_\tau^2)$ if it is random,

ω_k are constants and sum up to 0 if the row effect is fixed,
and are iid $N(0, \sigma_\omega^2)$ if it is random,

Ψ_l are constants and sum up to 0 if the Greek letter effect is fixed,
and are iid $N(0, \sigma_\psi^2)$ if it is random,

ε_{ijk} are iid $N(0, \sigma_\epsilon^2)$ (and independent of other effect if they are random).

Fitted Model

The least square estimators of the parameters in Graeco-Latin square model are:

Parameter Estimator

$$\mu \quad \bar{Y}_{....}$$

$$\theta_i \quad \bar{Y}_{i...} - \bar{Y}_{....}$$

$$\tau_j \quad \bar{Y}_{.j..} - \bar{Y}_{....}$$

$$\omega_k \quad \bar{Y}_{..k.} - \bar{Y}_{....}$$

$$\Psi_l \quad \bar{Y}_{...l} - \bar{Y}_{....}$$

The fitted values are:

$$\hat{Y}_{ijkml} = \bar{Y}_{i...} + \bar{Y}_{.j..} + \bar{Y}_{..k.} + \bar{Y}_{...l} - 3\bar{Y}_{....}$$

and the residuals are:

$$e_{ijklm} = Y_{ijklm} - \bar{Y}_{i...} - \bar{Y}_{.j..} - \bar{Y}_{..k.} - \bar{Y}_{...l} + 3\bar{Y}_{....}$$

We now proceed to discuss the ANOVA for the Graeco-Latin Square design and examine the SAS output for our data set.